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Air Quality Benefits Assessment Tool

Appendices

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Appendix A - Detailed Health Impact Calculations

This appendix details the formulas and algorithms used to calculate health impact estimates from the various input factors: concentration response functions, final concentration changes, baseline health endpoint occurrence rates, population counts, endpoint valuations and various global constants.

The purpose of this appendix is to comprehensively present all calculation details in an explicit and unambiguous manner, so that the more mathematically inclined user can fully understand how any output is derived and is also able to utilize the information to independently recreate a simulation. This purpose is not compatible with an easy to read, non-cryptic and simple to follow presentation, but we believe that the "painful" details need to be included somewhere.

This discussion is a sequential build-up of all the formulas implemented by AQBAT in generating outputs. Terminology and symbols are introduced as they are needed, and are then utilized anytime after their point of introduction.

First we need to set up the terminology and symbols used to build the formula for the estimate of the gross number of excess endpoint incidents attributable to a pollutant concentration change. This number is assigned to every specific <pollutant, endpoint, geographic area, scenario year> 4-tuple, where a concentration response function (CRF) is assigned to the corresponding <pollutant, endpoint, geographic area> triplet in your scenario model.

Let $n\%$ denote the general n^{th} percentile distribution concentration where:

$$n = 0, 1, 2, 3, 4, 5, 10, 15, 20, \dots, 95, 96, 97, 98, 99, 100,$$

and let mean denote the general mean of a distribution. Recall that the mean is always calculated as the weighted average of the percentile concentrations with weights assigned as (**%ile=weight**):

$$0\%=0.005, 1\%=0.01, 2\%=0.01, 3\%=0.01, 4\%=0.01, 5\%=0.03, 10\%=0.05, 15\%=0.05, 20\%=0.05, \dots, 90\%=0.05, 95\%=0.03, 96\%=0.01, 97\%=0.01, 98\%=0.01, 99\%=0.01, 100\%=0.005$$

Let this set of weights be designated as:

$$wt = \{wt_{0\%}, wt_{1\%}, \dots, wt_{5\%}, wt_{10\%}, \dots, wt_{95\%}, wt_{96\%}, \dots, wt_{100\%}\}.$$

Pollutant concentrations are dealt with as sets of distribution values in AQBAT. Let C denote a general set of concentration values. Then:

$$C = \{C_{mean}, C_{0\%}, C_{1\%}, C_{2\%}, C_{3\%}, C_{4\%}, C_{5\%}, C_{10\%}, C_{15\%}, \dots, C_{95\%}, C_{96\%}, C_{97\%}, C_{98\%}, C_{99\%}, C_{100\%}\}$$

where we adopt the convention that the symbol attached to C identifies the specific distribution point (mean or percentile) defining the concentration value; i.e. C_{mean} is the mean concentration value and $C_{n\%}$ is the $n\%$ ile concentration value, for the distribution of concentration values.

Note that such a set of concentration distribution values is for a specific <pollutant, averaging period, geographic area, scenario year> 4-tuple.

More specifically, let

$$SQ = \{SQ_{mean}, SQ_{0\%}, SQ_{2\%}, \dots, SQ_{5\%}, SQ_{10\%}, \dots, SQ_{95\%}, SQ_{96\%}, \dots, SQ_{100\%}\}$$

denote the status quo concentration distribution set, and let

$$FC = \{FC_{mean}, FC_{0\%}, FC_{2\%}, \dots, FC_{5\%}, FC_{10\%}, \dots, FC_{95\%}, FC_{96\%}, \dots, FC_{100\%}\}$$

denote the forecast concentration distribution set for the <pollutant, geographic area, scenario year> triplet.

Let CRF_{smp} denote the sampled β value / mean or pre-determined percentile value from the CRF distribution of values for the <pollutant, endpoint, geographic area> triplet. The discussion assumes that the averaging period for the pollutant concentration values defined above coincides with the averaging period for the CRF.

Let thr denote the threshold concentration assigned to the CRF.

Let eps denote the value 0.0001, the pre-set lowest non-zero concentration for each pollutant and averaging period, with respect to its unit of concentration measure, in the current version of AQBAT. This is a sufficiently small positive number chosen arbitrarily to replace zero divisors and arguments to the logarithm function.

Let br denote the endpoint baseline rate for the <endpoint, geographic area, scenario year> triplet. This number is an estimate of the number of incidents of the endpoint occurring for every million specified (or subject) population in the geographic area for the scenario year. This number may also depend on the population projection selected for the scenario. AQBAT acquires br from the health endpoint database, or from the CRF database, if this was entered on the **Concentration Response Function** form.

Let pop denote the specified population count for the <endpoint, geographic area, scenario year> triplet. The endpoint determines the specified population (specific proportion of an age group in the geographic area). This number may also depend on the population projection selected for the scenario.

Let bc denote the annual endpoint baseline counts for the <endpoint, geographic area, scenario year> triple. Then:

$$bc = \frac{br}{1,000,000} \times pop$$

Let paf denote the pollutant adjustment factor. Then

for a short-term or acute exposure health endpoint:

$$\text{for O}_3 \text{ (May-Sep),} \quad paf = \frac{153}{365} \quad \text{and}$$

$$\text{for all other pollutants,} \quad paf = 1.$$

for a long-term or chronic exposure health endpoint

$$\text{for all pollutants,} \quad paf = 1.$$

To explain these adjustment factors:

For a short-term or acute exposure CRF, a pollutant is associated with a health endpoint on a day to day basis, with **no** “memory” of the long term. The warm season O₃ associations therefore apply only to the corresponding days.

For a long-term or chronic exposure CRF, a pollutant is associated with a health endpoint on a long term basis, i.e. with “memory” of the long term. Hence even the warm season O₃ associations therefore apply to all days of the year. The epidemiology is asserting that the warm season is representing the whole year, even though data was not considered/collected for this entire period.

Let $PctXS$ denote the % excess adverse endpoints for the final pollutant concentration change from status quo concentration to forecast concentration (**note: 1% = 0.01**).

If **either** the endpoint is a long-term or chronic exposure endpoint, or the threshold concentration is zero, then:

for a log linear regression type,

$$PctXS = 1 - \exp \{ -CRF_{smp1} \times [\max(FC_{mean}, thr) - \max(SQ_{mean}, thr)] \}$$

for a linear regression type,

$$PctXS = CRF_{smp1} \times [\max(FC_{mean}, thr) - \max(SQ_{mean}, thr)]$$

and for a **Burnett non-linear** regression type (employed only for a chronic exposure type mortality),

$$PctXS = 1 - \exp (-CRF_{smp1} \times (T(\max(FC_{mean} - thr, 0)) - T(\max(SQ_{mean} - thr, 0))))$$

where

$$T(x) = \ln((x + \theta)/\theta) / (1 + \exp(-(x - \mu)/\pi))$$

is defined as the transform of general pollutant concentration x , with θ , μ and π the additional parameters required to define this regression type.

where $\max$ denotes the maximum value among the values listed between the corresponding parentheses, $\exp$ denotes the exponential (base e) function and $\ln$ denotes the natural logarithm function.

If the endpoint is a short-term exposure or acute health endpoint, **and** the threshold concentration is positive, then (with “*” and “+” denoting multiplication and addition, respectively):

$$PctXS = PctXS_{mean} =$$

$$wt_{0\%} * PctXS_{0\%} + wt_{1\%} * PctXS_{1\%} + \dots + wt_{5\%} * PctXS_{5\%} + wt_{10\%} * PctXS_{10\%} + \dots + wt_{95\%} * PctXS_{95\%} + wt_{96\%} * PctXS_{96\%} + \dots + wt_{100\%} * PctXS_{100\%}$$

where, for $m = 0, 1, 2, \dots, 5, 10, 15, \dots, 95, 96, 97, \dots, 100$

for a log linear regression type,

$$PctXS_{m\%} = 1 - \exp \{ -CRF_{smp1} \times [\max(FC_{m\%}, thr) - \max(SQ_{m\%}, thr)] \}$$

for a linear regression type,

$$PctXS_{m\%} = CRF_{smp1} \times [\max(FC_{m\%}, thr) - \max(SQ_{m\%}, thr)]$$

What is going on here is that we are calculating the **effective** % excess, taking into account the threshold concentration. AQBAT assumes that there are no health associations below the threshold concentration. For a long-term or chronic exposure CRF, or a zero threshold concentration, it suffices to adjust the status quo and forecast mean concentrations. For a short-term or acute exposure CRF with positive threshold concentration, one has to make the adjustments at the distribution level, where, in this case, each percentile concentration should be viewed as representing a specific % of days, reflected by the weight assigned to the %ile, at the corresponding concentration.

To this point, we have determined **PctXS**, the effective % excess adverse endpoints.

For a **chronic exposure mortality type health endpoint and only for the last scenario year**, the life years gained estimate, **LYG**, is a degree 2 polynomial of the **PctXS**.

Let $AbsPctXS = \min(|PctXS|, 10)$, where $||$ denotes the absolute value operator. $AbsPctXS$ should not exceed 10 = 1000%.

From the table of degree 2 polynomial coefficients, we can extract a specific intercept, a degree 1 coefficient (coeff1) and a degree 2 coefficient (coeff2), depending on the mortality type and which of 6 ranges that $AbsPctXS$ lies in.

Then, applying the specific coefficients found,

$$\begin{aligned} LYG &= \text{intercept} + (\text{coeff1} \times AbsPctXS) + (\text{coeff2} \times AbsPctXS \times AbsPctXS) \\ LYG &= -1 \times LYG, \text{ if } PctXS > 0. \end{aligned}$$

Continuing, with these constituent factors set up,

letting **GC** denote the estimate of the gross number of excess endpoint incidents attributable to a pollutant concentration change, assigned to the specific <pollutant, endpoint, geographic area, scenario year> 4-tuple,

we obtain

$$GC = PctXS \times bc \times paf$$

This is the sample count estimate for the specific <pollutant, endpoint, geographic area, scenario year> 4-tuple, resulting from one iteration's sample CRF_{smp} in the simulation / or one mean or pre-determined percentile.

For example GC refers to the more completely specified $GC<\text{pollutant, endpoint, geographic area, scenario year}>$.

From here on, we include the n-tuple index that the acronym refers to specifically, whenever necessary in order to remain unambiguous.

Let **AGC** denote the adjusted gross count, and **NC** denote the netted count, for the 4-tuple specified above. The netted count is needed to calculate the \$ valuation estimate. The adjusted gross count is an intermediate result used to derive the netted (net) count.

If the corresponding CRF is netted then

$$AGC = GC$$

Otherwise (CRF not already netted), as a general formula:

$$\begin{aligned} &AGC<\text{pollutant, endpoint, geographic area, scenario year}> \\ &= \\ &GC<\text{pollutant, endpoint, geographic area, scenario year}> \end{aligned}$$

The netted endpoints {corresponding netting endpoints} are:

- Acute Respiratory Symptom Days
{Minor Restricted Activity Days, Restricted Activity Days},
- Cardiac Emergency Room Visits
{Cardiac Hospital Admissions, Elderly Cardiac Hospital Admissions}, and
- Respiratory Emergency Room Visits
{Respiratory Hospital Admissions}.

AQBAT enforces the convention that the **netted count must end up between zero and the gross count**, based on the following reasoning.

During a simulation, the possibility exists that, for example, if Respiratory Hospital Admissions (RHAs) are to be netted from Respiratory Emergency Room Visits (RERVs), for an iteration, corresponding CRF_{smpl} for <pollutant, RERV> and <pollutant, RHA> may result in a higher estimated RHA gross count than RERV gross count, or even counts of opposite sign. We still want the initial <pollutant, RERV> CRF_{smpl} and resulting count estimate to be the prevalent one, and so we enforce this rule.

The implementation is as follows:

if $GC < 0$ then	$NC = \min\{\max(GC, AGC), 0\}$,
if $GC = 0$ then	$NC = 0$, and
if $GC > 0$ then	$NC = \max\{\min(GC, AGC), 0\}$,

where $\min$ and $\max$ denote the minimum and maximum values, respectively, among the values listed between corresponding parentheses.

So far we have derived the gross count (GC) and net count (NC) estimates for a <pollutant, endpoint, geographic area, scenario year> 4-tuple, where the <pollutant, endpoint, geographic area> triple has been assigned a CRF, and the scenario year is not an "in-between" year.

We have also initially derived the life years gained (LYG) estimates for chronic exposure mortality type health endpoints and the last scenario year (degree 2 polynomial of $PctXS$).

These estimates correspond to a pollutant concentration change from the status quo concentration to the forecast concentration for the geographic area and scenario year.

There are two other output types, proportion of health baseline and \$ valuations, corresponding to the counts. Thus we need to derive $G\%$ and $N\$$ which denote the proportion of health baseline for the gross count and \$ valuation for the net count, respectively, for the <pollutant, endpoint, geographic area, scenario year> 4-tuple.

For the proportion of health baseline:

$$G\% = GC / bc$$

The derivation $N\$$ is more involved and discussed below.

Let EPV_{smpl} denote the sampled estimate value / pre-determined mean or percentile from the distribution of values defined for the Endpoint Valuation (EPV) assigned to the <endpoint, geographic area> pair. We want to convert this value to the selected **currency year** Canadian \$s after also applying the inputted **discount rate** and selected **base year for discounting**.

For the EPV assigned to the <endpoint, geographic area> pair, let

- IT denote the indexing source type (one of CPI or MCI) and
- SY denote the source year.

In addition, as global settings, let:

- CY denote the currency year,
- BYD denote the base year for discounting and
- DR denote the discount rate.

The three terms above are standard terms in the field of Economics.

We list here the Price Index (PI) table of conversion factors to apply:

Year	CPI	MCI
1996	100.0	100.0
1997	101.6	101.4
1998	102.5	103.7
1999	104.3	106.0
2000	107.2	108.4
2001	109.9	111.3
2002	112.4	113.3
2003	115.5	115.4
2004	117.7	117.5
2005	120.2	120.6
2006	122.6	123.1
2007	125.2	125.6
2008	128.2	127.9
2009	128.6	131.4
2010	131.0	134.3
2011	134.8	136.0
2012	136.8	136.8
2013	138.1	136.6
2014	140.8	137.8
2015	142.4	139.5
2016	144.4	141.3
2017	146.6	143.6

We then use the Price Index (PI) table, above, indexed by currency year and indexing type, to obtain the monetary conversion factor, denoted by *mcf*, as follows:

$$mcf = \frac{\left[\frac{PI(CY, IT)}{PI(SY, IT)} \right]}{\left[1 + DR \right]^{(ScenarioYear - BYD)}}$$

With the above factors defined and derived, we obtain:

$$N\$ = NC \times EPV_{smp} \times mcf$$

We have thus derived *GC*, *G%*, *LYG* (if applicable, **as noted above**) and *N\$* for every <pollutant, endpoint, geographic area, scenario year> 4-tuple, where the scenario year is not an "in-between" year, such that a CRF has been assigned to the <pollutant, endpoint, geographic area> triple in your scenario model. We call each such 4-tuple a fundamental 4-tuple.

We have thus derived formulas for any AQBAT output for your scenario model. With these formulas you can recreate all central estimates from the central CRF beta values, the central EPV estimate values, the status quo and forecast concentrations, the global scenario settings, population counts and baseline endpoint rates.

Sampling from the CRF and EPV distributions, you can recreate the Monte Carlo simulation to track these outputs and generate distribution statistics for them.

Health Endpoint Listing with Characteristics

We list below the exposure type (acute or chronic) and specified population for the 18 fixed AQBAT health endpoints. These characteristics are pre-defined in AQBAT; the user cannot change them.

1. Acute Exposure Mortality -
acute, 100% of the total population (all ages),
2. Chronic Exposure Mortality -
chronic, 100% of the population 25 and over,
3. Chronic Exposure Cerebrovascular Mortality -
chronic, 100% of the population 25 and over,
4. Chronic Exposure COPD (chronic obstructive pulmonary disease) Mortality -
chronic, 100% of the population 25 and over,
5. Chronic Exposure Ischemic Heart Disease Mortality -
chronic, 100% of the population 25 and over,
6. Chronic Exposure Lung Cancer Mortality -
chronic, 100% of the population 25 and over,
7. Chronic Exposure Respiratory Mortality -
chronic, 100% of the population 30 and over,
8. Acute Respiratory Symptom Days -
acute, 100% of all adults (20 and over) and 82.3% (non-asthmatic) of children aged 5 to 19,
9. Adult Chronic Bronchitis Cases -
chronic, 100% of the population 25 and over,
10. Asthma Symptom Days -
acute, 17.7% (asthmatic) of children aged 5 to 19,
11. Cardiac Emergency Room Visits -
acute, 100% of the total population (all ages),
12. Cardiac Hospital Admissions -
acute, 100% of the total population (all ages),
13. Child Acute Bronchitis Episodes -
chronic, 100% of children aged 5 to 19,
14. Elderly Cardiac Hospital Admissions -
acute, 100% of the population 65 and over,
15. Minor Restricted Activity Days -
acute, 100% of all adults (20 and over) and 82.3% (non-asthmatic) of children aged 5 to 19,
16. Respiratory Emergency Room Visits -
acute, 100% of the total population (all ages),

17. Respiratory Hospital Admissions -
acute, 100% of the total population (all ages), and
18. Restricted Activity Days -
acute, 100% of all adults (20 and over) and 82.3% (non-asthmatic) of children aged 5 to 19.

Appendix B – Health Canada Endorsed Concentration Response Functions and Endpoint Valuations

Concentration Response Functions

Gases (CO, NO₂, O₃ and SO₂) and Acute Exposure Mortality

Burnett RT, Stieb D, Brook JR, Cakmak S, Dales R, Raizenne M, Vincent R, Dann T. Associations between short-term changes in nitrogen dioxide and mortality in Canadian cities. Arch Environ Health. 2004 May;59(5):228-36. - result from model with 4 gases provided by 1st author, additional to published results.

Burnett et al. (2004) reported on an analysis of air pollution and mortality in 12 Canadian cities. The author provided results from additional multi-pollutant models not reported in the paper; the 4-gas model was selected based on the overall t-value among the candidate models. Percent excess mortality (associated with the mean pollutant concentration) from Poisson regression models for CO, NO₂, O₃ and SO₂ respectively were: 0.19% (T = 0.73, 1.0 ppm), 1.69% (T = 3.00, 22.4 ppb), 2.60% (T = 6.16, 30.6 ppb), and 0.23% (T = 2.09, 5.0 ppb). These results translate into regression coefficients (standard errors) of: CO, 0.00190 (0.00260), NO₂, 0.000748 (0.000249), O₃, 0.000839 (0.000136), and SO₂, 0.000459 (0.000220). Although this multi-pollutant model excluded particulate matter, it was selected as the model which best reflected the impact of the overall air pollution mix. Because of multi-collinearity among pollutants, this model should nonetheless still reflect impacts of particulate matter. In any case, the effects of particulate matter in this study were reduced substantially when modelled together with NO₂, the effect of which predominated in this analysis.

O₃ (May-Sep) and Chronic Exposure Respiratory Mortality

Jerrett M, Burnett RT, Pope CA III, Ito K, Thurston G, Krewski D, Shi Y, Calle E, Thun M. Long-term ozone exposure and mortality. N Engl J Med. 2009 Mar 12;360(11):1085-95.

Jerrett et al. (2009) analyzed data from the American Cancer Society cohort study. The relative risk of death from respiratory causes was 1.040 (95% confidence interval, 1.010 to 1.067) per 10 ppb ozone in a model with PM_{2.5}; exposure was based on average of quarterly averages with $\geq 75\%$ of daily values. This translates into a coefficient of 0.00392 with standard error 0.00132.

PM_{2.5} and Chronic Exposure Chronic Obstructive Pulmonary Disease (COPD), Cerebrovascular and Ischemic Heart Disease

Shin HH, Cohen AJ, Pope CA 3rd, Ezzati M, Lim SS, Hubbell BJ, Burnett RT (2016). Meta-Analysis Methods to Estimate the Shape and Uncertainty in the Association Between Long-Term Exposure to Ambient Fine Particulate Matter and Cause-Specific Mortality Over the Global Concentration Range. Risk Anal. 36(9): 1813-1825.

Cause-specific CRFs were derived from meta-analyses of worldwide cohort studies as described in Shin et al. 2013. The values specified for beta and standard error in AQBAT are actually the alpha and beta (shape and scale) parameters of the gamma distribution (see table below). The mean of the pooled hazard ratio can be calculated as the exponential of the product of the shape and scale parameters of the gamma distribution, while the 95% confidence interval can be calculated as the exponential of the inverse of the gamma distribution at the 2.5th and 97.5th percentiles of the cumulative density function.

Estimates of shape (*a*) and scale (*b*) of gamma distributions representing variation in risk of mortality among cohort studies associated with a 10 µg/m³ change in PM_{2.5}

Health endpoint	Hazard ratio (95% CI) based on 10 µg/m ³ change in PM _{2.5} – reference ^{a,b}	Gamma distribution <i>G(a, b)</i> ^c
Chronic exposure cerebrovascular mortality	1.12 (1.01, 1.24) – ACS 0.89 (0.67, 1.18) – SC 1.83 (1.11, 3.00) – WHI 1.62 (1.07, 2.44) – DSDC 1.16 (0.92, 1.46) – CT	<i>G</i> (4.884, 0.03375)
Chronic exposure COPD mortality	1.05 (0.95, 1.17) – ACS 1.17 (0.85, 1.62) – SC 1.21 (0.88, 1.68) – CT	<i>G</i> (14.57, 0.00601)
Chronic exposure ischemic heart disease mortality	1.26 (1.16, 1.38) – ACS 1.33 (1.16, 1.52) – SC 2.21 (1.17, 4.16) – WHI 2.02 (1.07, 3.78) – NH 1.00 (0.87, 1.15) – ASHS 0.96 (0.75, 1.22) – DSDC 1.20 (1.02, 1.41) – CT 0.98 (0.71, 1.36) – MHP	<i>G</i> (1.156, 0.2117)

CI: confidence interval; COPD: chronic obstructive pulmonary disease; PM_{2.5}: fine particulate matter or particulate matter with an aerodynamic diameter of 2.5 µm or less

^a References:

ACS (American Cancer Society): Krewski et al. (2009).

ASHS (Adventist Study of Health and Smog): Abbey et al. (1999) / Chen et al. (2005).

CT (California Teachers Study): Lipsett et al. (2011).

DSDC (Dutch Study of Diet and Cancer): Beelen et al. (2008).

MHP (Male Health Professionals' Study): Puett et al. (2011).

NH (Nurses' Health Study): Puett et al. (2008) / Puett et al. (2009).

SC (Six City Study): Laden et al. (2006).

WHI (Women's Health Initiative): Miller et al. (2007).

^b Hazard ratio converted to 1.00 for estimation of gamma distributions.

American Cancer Society Study (ACS):

Krewski, Jerrett M, Burnett RT, Ma R, Hughes E, Shi Y, Turner MC, Pope III CA, Thurston G, Calle EE, Thun MJ (2009) Extended follow-up and spatial analysis of the American Cancer Society study linking particulate air pollution and mortality. Health Effects Institute Report

Adventist Study of Health and Smog (AHSMOG)

Abbey DE, Nishino N, McDonnell WF, Burchette RJ, Knutsen SF, Beeson WL, Yang JX (1999) Long-Term Inhalable Particles and Other Air Pollutants Related to Mortality in Nonsmokers. American Journal of Respiratory and Critical Care Medicine 159:373–382.

Chen LH, Knutsen SF, Shavlik D, Beeson LW, Petersen F, Ghamsary M, Abbey D (2005). The Association between Fatal Coronary Heart Disease and Ambient Particulate Air Pollution: Are Females at Greater Risk? Environmental Health Perspectives 113:1723–1729.

California Teachers Study

Lipsett MJ, Ostro BD, Reynolds P, Goldberg D, Hertz A, Jerrett M, Smith DF, Garcia C, Chang ET, Bernstein L (2011). Long-term exposure to air pollution and cardiorespiratory disease in the California Teachers Study cohort. Am J Respir Crit Care Med 184(7): 828–835.

Dutch Study on Diet and Cancer (DSDC):

Beelen R, Hoek G, van den Brandt AP, Goldbohm RA, Fischer P, Schouten LJ, Jerrett M, Hughes E, Armstrong B, Brunekreef B (2008). Long-Term Effects of Traffic-Related Air Pollution on Mortality in a Dutch Cohort (NLCS-AIR Study). *Environmental Health Perspectives* 116:196–202.

Male Health Professionals Study

Puett RC, Hart JE, Suh H, Mittleman M, Laden F (2011). Particulate matter exposures, mortality, and cardiovascular disease in the Health Professionals Follow-up Study. *Environ Health Perspect* 119(8): 1130–1135.

Nurses Health Study (NHS):

Puett RC, Schwartz J, Hart JE, Yanosky JD, Speizer FE, Suh H, Paciorek CJ, Neas LM, Laden F (2008) Chronic Particulate Exposure, Mortality, and Coronary Heart Disease in the Nurses' Health Study. *American Journal of Epidemiology* 168:1161–1168.

Puett RC, Hart JE, Yanosky JD, Paciorek C, Schwartz J, Suh H, Speizer FE, Laden F (2009). Chronic Fine and Coarse Particulate Exposure, Mortality and Coronary Heart Disease in the Nurses' Health Study. *Environmental Health Perspectives* doi: 10.1289/ehp.0900572 (available at <http://dx.doi.org/>) Online 15 June 2008

Six Cities Study (SCS):

Laden F, Schwartz J, Speizer FE, Dockery DW (2006) Reduction in Fine Particulate Air Pollution and Mortality Extended Follow-up of the Harvard Six Cities Study. *American Journal of Respiratory and Critical Care Medicine* 173:667–672.

Women's Health Initiative (WHI):

Miller KA, Siscovick DS, Sheppard L, Shepherd K, Sullivan JH, Anderson GL, Kaufman JD (2007) Long-Term Exposure to Air Pollution and Incidence of Cardiovascular Events in Women. 356:447-58.

O₃ (May-Sep) and Acute Respiratory Symptom Days

Krupnick AJ, Harrington W, Ostro B. 1990. Ambient Ozone and Acute Health Effects: Evidence from Daily Data. *Journal of Environmental Economics and Management*, 18(1): 1-18.

Krupnick, Harrington and Ostro (1990) reported on the association between ozone and the occurrence of acute respiratory symptoms in a panel of California families. They employed a Markov model which accounted for the occurrence of symptoms on the previous day and adjusted for coefficient of haze, nitrogen dioxide and sulphur dioxide as co-pollutants. The incremental change in frequency of symptoms was calculated by substituting the coefficient from table V, column 3, divided by 10 to convert from pphm to ppb, together with the transitional probabilities, $p_1=0.775$ and $p_2=0.0468$ (provided by the authors), into equation 3 on page 12 of the paper. The baseline frequency of symptoms was calculated by substituting p_1 and p_0 into equation 2. Thus the proportional change per ppb ozone is the output from equation 3 divided by that of equation 2, 0.000786 (SE 0.000386).

PM_{2.5} and Acute Respiratory Symptom Days

Krupnick AJ, Harrington W, Ostro B. 1990. Ambient Ozone and Acute Health Effects: Evidence from Daily Data. *Journal of Environmental Economics and Management*, 18(1): 1-18.

Krupnick, Harrington and Ostro (1990) reported on the association between coefficient of haze (coh) and the occurrence of acute respiratory symptoms in a panel of California families. They employed a Markov model which accounted for the occurrence of symptoms on the previous day and adjusted for ozone, nitrogen dioxide and sulphur

dioxide as co-pollutants. The incremental change in frequency of symptoms was calculated by substituting the coefficient from table V, column 3, multiplied by 0.211 to convert from coh to PM_{2.5}, together with the transitional probabilities, p1=0.775 and p2=0.0468 (provided by the authors), into equation 3 on page 12 of the paper. The conversion from coh to PM_{2.5} was calculated by dividing the ratio of coh to TSP (0.116) provided by the authors, by that of PM₁₀ to TSP (0.55) provided by Environment Canada. This assumes that the toxicity of PM_{2.5} per µg/m³ is the same as that of PM₁₀. The baseline frequency of symptoms was calculated by substituting p1 and p0 into equation 2. Thus the proportional change per µg/m³ PM_{2.5} is the output from equation 3 divided by that of equation 2, 0.00266 (SE 0.00139).

PM_{2.5} and Adult Chronic Bronchitis Cases

Abbey DE, Lebowitz MD, Mills PK, Petersen FF, Beeson WL, Burchette RJ. 1995. Long term Ambient Concentrations of Particulates and Oxidants and Development of Chronic Disease in a Cohort of Nonsmoking California Residents. *Inhalation Toxicology*, 7:19-34.

Abbey et al. (1995) reported the results of a cohort study of air pollution and the development of chronic lung disease among non-smoking Seventh Day Adventists living in California. Based on a logistic regression model, which also included personal characteristics, they reported an odds ratio of 1.81 (95%CI 0.98-3.25) for the development of chronic bronchitis per 45 µg/m³ PM_{2.5} (table 2, row 2). Taking the natural log of the odds ratio and dividing by 45 yields a coefficient of 0.0132 (SE 0.00680) per µg/m³ PM_{2.5}. They reported 10 year incidence of chronic bronchitis was 6.26% (117 new cases occurred among 1868 subjects for whom PM_{2.5} exposures could be estimated). We calculate the annual incidence, p1, from the expression: $0.0626 = 1 - (1 - p1)^{10}$, so that p1=0.00644.

O₃ (May-Sep) and Asthma Symptom Days

Mortimer KM, Neas LM, Dockery DW, Redline S, Tager IB. 2002. The effect of air pollution on inner-city children with asthma. *Eur Respir J*, 19:699–705.

Schildcrout JS, Sheppard L, Lumley T, Slaughter JC, Koenig JQ, Shapiro GG. 2006. Ambient air pollution and asthma exacerbations in children: an eight-city analysis. *Am J Epidemiol* 164:505–517.

Numerous panel studies have been conducted on the association between O₃ and asthma exacerbations in children. Several of these were carried out in summer camps, which may not reflect typical exposure conditions, in that campers would be expected to spend more time outdoors compared with non-campers. Others have been conducted in locations such as Mexico City and Los Angeles, which experience very high O₃ concentrations not representative of conditions in Canada. We therefore selected two large multicentre North American panel studies as the source of the CRF. Mortimer et al. (2002) analyzed data collected in summer 1993 for 846 inner-city children aged 4–9 years from eight American cities. The average 8 h maximum O₃ concentration among all cities was 48 ppb. The odds ratio for morning asthma symptoms was 1.16 (95% CI 1.02–1.30) in relation to a 15 ppb increment on average in lag 1–5 day O₃. This was reduced to 1.07 (0.92–1.26) in a joint model with NO₂ in seven cities and to 1.04 (0.70–1.55) in a joint model with PM₁₀ based on three cities (table 4). Schildcrout et al. (2006) analyzed data collected from 1993 to 1995 for 990 children aged 5–13 years, also from eight cities including Toronto, and only with Baltimore in common with the Mortimer et al. (2002) analysis. Median 1 h maximum O₃ concentrations ranged from 43 to 65.8 ppb. The odds ratio for asthma symptoms was 1.06 (95% CI 0.92–1.23) in relation to a 30 ppb increment in lag 0 O₃ (the largest effect among lags considered; figure 1). Joint models with other pollutants were not run. The log odds ratio from Mortimer et al. (2002) based on the 8 h maximum (joint model with NO₂) was multiplied by 1.13 (the ratio of 1 h maximum to 8 h maximum in Canadian cities) and pooled with the Schildcrout et al. (2006) result to obtain an odds ratio of 1.05 (95% CI 0.96–1.14) per 20 ppb. The same baseline frequency of asthma symptoms and prevalence of current wheeze as for PM_{2.5} were applied to 14.3% of children aged 5–19 years.

PM_{2.5} and Asthma Symptom Days

Dell SD, Foty RG, Gilbert NL, Jerrett M, To T, Walter SD, Stieb DM. 2010. Asthma and allergic disease prevalence in a diverse sample of Toronto school children: results from the Toronto Child Health Evaluation Questionnaire (T-CHEQ) Study. *Can Respir J* 171:e1–6.

Mortimer KM, Neas LM, Dockery DW, Redline S, Tager IB. 2002. The effect of air pollution on inner-city children with asthma. *Eur Respir J* 19:699–705.

Ward DJ, Ayres JG. 2004. Particulate air pollution and panel studies in children: a systematic review. *Occup Environ Med* 61:e13 (doi:10.1136/oem.2003.007088).

Weinmayr G, Romeo E, De Sario M, Weiland SK, Forastiere F. 2010. Short-term effects of PM₁₀ and NO₂ on respiratory health among children with asthma or asthma-like symptoms: a systematic review and meta-analysis. *Environ Health Perspect* 118:449–457.

These parameters are derived using the same approach as described in the Health Risk of Air Pollution in Europe project of the WHO European Centre for Environment and Health. Weinmayr et al. (2010) conducted a systematic review and meta-analysis based on 36 studies of the association between air pollution and asthma symptoms in children. The pooled odds ratio was 1.028 (95% CI 1.006–1.051) per 10 µg/m³ PM₁₀ (table 2) based on a random effect model including all studies. This is based on single pollutant models, as results from multi-pollutant models were not consistently available. However, the derived effect size is nonetheless much smaller than that observed in a multi-pollutant model for North American cities in Mortimer et al. (2002). In order to derive an odds ratio for PM_{2.5}, we multiplied the log odds ratio for PM₁₀ by 2.37, which is the average of the ratio of log pooled odds ratios for PM_{2.5} vs. PM₁₀ for cough and other respiratory symptoms reported by Ward and Ayres (2004; tables 3 and 4) in their earlier meta-analysis. The result is an odds ratio for PM_{2.5} of 1.07 (95% CI 1.01–1.12). The baseline daily frequency of asthma symptoms in asthmatic children varies widely in panel studies. We have conservatively estimated it at 20%. The population to which this is applicable is based on the prevalence of current wheeze in Canada from the National Longitudinal Survey of Children and Youth (14.3%; Dell et al. 2010).

PM_{2.5} and Cardiac Emergency Room Visits

Burnett, RT, Dales R, Krewski D, Vincent R, Dann T, Brook JR. 1995. Associations between Ambient Particulate Sulphate and Admissions to Ontario Hospitals for Cardiac and Respiratory Diseases. *American Journal of Epidemiology*, 142(1): 15-22.

Stieb DM, Beveridge RC, Brook JR, Smith-Doiron M, Burnett RT, Dales RE, Beaulieu S, Judek S, Mamedov A. Air pollution, aeroallergens and cardiorespiratory emergency department visits in Saint John, Canada. *J Expo Anal Environ Epidemiol*. 2000 Sep-Oct;10(5):461-77.

Substantially more data are available pertaining to air pollution and hospital admissions in Canada relative to emergency department visits. We therefore elected to represent the effects of air pollution on cardiac emergency department visits using the results for hospital admissions scaled up in number based on the relative frequency of hospital admissions and emergency visits for these conditions. Thus, the change in frequency per unit air pollution was the same as for hospital admissions based on Burnett et al. (1995) i.e. 0.0711 % (SE 0.0170 %) increase per 1 µg/m³. The baseline rate of emergency visits is equal to the baseline rate of hospital admissions divided by 0.760, the proportion of visits resulting in hospital admission as reported by Stieb et al. (2000).

PM_{2.5} and Cardiac Hospital Admissions

Burnett, RT, Dales R, Krewski D, Vincent R, Dann T, Brook JR. 1995. Associations between Ambient Particulate Sulphate and Admissions to Ontario Hospitals for Cardiac and Respiratory Diseases. *American Journal of Epidemiology*, 142(1): 15-22.

Burnett et al. (1995) reported a 3.3 % (95%CI 1.7-4.8 %) increase in cardiac hospital admissions per $13 \mu\text{g}/\text{m}^3$ sulfate based on a linear regression model which also included ozone and temperature (table 5, row 2). Multiplying by the average ratio of sulfate to $\text{PM}_{2.5}$ of 0.28 (Environment Canada), this equates to a 0.0711 % (SE 0.0170 %) increase per $1 \mu\text{g}/\text{m}^3 \text{PM}_{2.5}$.

$\text{PM}_{2.5}$ and Child Acute Bronchitis Episodes

Dockery, DW, Cunningham J, Damokosh AI, Neas LM, Spengler JD, Koutrakis P, Ware JH, Raizenne M, Speitzer FE. 1996. Health Effects of Acid Aerosols on North American Children: Respiratory Symptoms. *Environmental Health Perspectives*, 104:500-505.

Hoek G, Pattenden S, Willers S, Antova T, Fabianova E, Braun-Fahrlander C, Forastiere F, Gehring U, Luttmann-Gibson H, Grize L, Heinrich J, Houthuijs D, Janssen N, Katsnelson B, Kosheleva A, Moshhammer H, Neuberger M, Privalova L, Rudnai P, Speizer F, Slachtova H, Tomaskova H, Zlotkowska R, Fletcher T. 2012. PM_{10} , and children's respiratory symptoms and lung function in the PATY study. *Eur Respir J* 403:538–547.

These parameters are derived using the same approach as described in the Health Risk of Air Pollution in Europe project of the WHO European Centre for Environment and Health. Hoek et al. (2012) conducted a meta-analysis of eight cross-sectional studies from Europe and North America, including the 24 cities study, which included data from several Canadian communities. The random effect pooled estimate of the odds ratio was 1.08 (95% CI 0.98–1.19) per $10 \mu\text{g}/\text{m}^3 \text{PM}_{10}$ (table 3), adjusted for age, sex, maternal education, paternal education, household crowding, current parental smoking, mother smoking during pregnancy, gas cooking, unvented gas/oil/kerosene heater, mould, nationality, birth order and “ever had a pet.” The effect size was reduced based on joint models with SO_2 , but this was based on only three studies (table 4). The average prevalence of bronchitis among the studies was 18.6% (table 2). In the 24 cities study (Dockery et al. 1996), the odds ratio for bronchitis for $\text{PM}_{2.5}$ was identical to that for PM_{10} across the exposure difference between highest- and lowest-exposure communities, 17.3 and $14.9 \mu\text{g}/\text{m}^3$ for PM_{10} and $\text{PM}_{2.5}$, respectively (tables 1 and 4). We therefore multiply the log of the pooled odds ratios for PM_{10} by this ratio (1.16) in order to derive a log odds ratio per $10 \mu\text{g}/\text{m}^3 \text{PM}_{2.5}$, resulting in an odds ratio of 1.09 (95% CI 0.98–1.22)..

$\text{PM}_{2.5}$ and Chronic Exposure Internal Cause Mortality

Crouse DL, Peters PA, van Donkelaar A, Goldberg MS, Villeneuve PJ, Brion O, Khan S, Atari DO, Jerrett M, Pope CA, Brauer M, Brook JR, Martin RV, Stieb D, Burnett RT. Risk of nonaccidental and cardiovascular mortality in relation to long-term exposure to low concentrations of fine particulate matter: a Canadian national-level cohort study. *Environ Health Perspect*. 2012 May;120(5):708-14. doi: 10.1289/ehp.1104049.

Crouse et al. (2012) examined the association between $\text{PM}_{2.5}$ derived from satellite observations and mortality during ten years of follow-up of a cohort of 2.1 million Canadians based on the 1991 long form census. Using a spatial random-effects Cox model including individual and ecological covariates and an urban/rural indicator, and accounting for spatial autocorrelation among cohort members, they reported a hazard ratio of 1.10 (95% CI 1.05-1.15) per $10 \mu\text{g}/\text{m}^3 \text{PM}_{2.5}$. This translates to a β of 0.00953 with standard error 0.00232.

CO and Elderly Cardiac Hospital Admissions

Burnett RT, Dales RE, Brook JR, Raizenne ME, Krewski D. Association between ambient carbon monoxide levels and hospitalizations for congestive heart failure in the elderly in 10 Canadian cities. *Epidemiology*. 1997 Mar;8(2):162-7.

Schwartz J and Morris R. 1995. Air Pollution and Hospital Admissions for Cardiovascular Disease in Detroit, Michigan. *American Journal of Epidemiology*, 142: 23-35.

Burnett et al. (1997) reported a 6.1% (T-ratio = 3.5) increase in hospital admissions for congestive heart failure among individuals over 65 years of age for an increase from 1 to 3 ppm daily one hour maximum CO (table 4, row

1). This was from a Poisson regression model with $\ln(\text{CO})$, also controlling for NO_2 , SO_2 , O_3 , coefficient of haze (COH), temperature and dewpoint temperature. Schwartz and Morris (1995) reported a 2.2% (95% CI 1.0-3.4%) increase in hospital admissions for congestive heart failure in those 65 and over in a Poisson regression model with PM_{10} , temperature and humidity per 1.28 ppm one hour maximum CO (table 6, row 6). These two estimates were pooled as percent increase in hospital admissions per ppm one hour maximum CO, using a random effects model. Because the Burnett et al. result was based on $\ln(\text{CO})$, the percent increase depends on the baseline concentration. We set this at 1 ppm as being representative of conditions in Canadian cities, thus the Burnett et al. result was expressed as a 4.32% (SE 1.23%) increase in admissions for a change from 1 to 2 ppm and that from Schwartz and Morris as 1.71% (SE 0.472%). The final pooled estimate was 2.77% (SE 1.28%).

O_3 (May-Sep) and Minor Restricted Activity Days

Ostro BD and Rothschild S. 1989. Air Pollution and Acute Respiratory Morbidity: An Observational Study of Multiple Pollutants. *Environmental Research*, 50:238-247.

Ostro and Rothschild (1989) reported an association between ozone and minor reduced activity days based on an analysis of data from the U.S. Health Interview Survey. They reported results by year for 1976-1981 based on a Poisson regression model including both ozone and $\text{PM}_{2.5}$ (table 4, column 2). We pooled these coefficients using a random effects model and obtained a pooled estimate of 0.000530 (SE 0.00291) per ppb daily 1 hour maximum ozone. The baseline daily rate of MRADs per person was $7.8/365 = 0.0214$.

O_3 (May-Sep) and Respiratory Emergency Room Visits

Burnett RT, Brook JR, Yung WT, Dales RE, Krewski D. 1997. Association between Ozone and Hospitalization for Respiratory Diseases in 16 Canadian Cities. *Environmental Research*, 72: 24–31.

Stieb DM, Beveridge RC, Brook JR, Smith-Doiron M, Burnett RT, Dales RE, Beaulieu S, Judek S, Mamedov A. Air pollution, aeroallergens and cardiorespiratory emergency department visits in Saint John, Canada. *J Expo Anal Environ Epidemiol*. 2000 Sep-Oct;10(5):461-77.

Substantially more data are available pertaining to air pollution and hospital admissions in Canada relative to emergency department visits. We therefore elected to represent the effects of air pollution on respiratory emergency department visits using the results for hospital admissions scaled up in number based on the relative frequency of hospital admissions and emergency visits for these conditions. Thus, the coefficient per unit air pollution was the same as for hospital admissions based on Burnett et al. (1997) i.e. 0.000791 (SE 0.000355) per ppb. The baseline rate of emergency visits is equal to the baseline rate of hospital admissions divided by 0.198, the proportion of visits resulting in hospital admission as reported by Stieb et al. (2000).

$\text{PM}_{2.5}$ and Respiratory Emergency Room Visits

Burnett RT, Dales R, Krewski D, Vincent R, Dann T, Brook JR. 1995. Associations between Ambient Particulate Sulphate and Admissions to Ontario Hospitals for Cardiac and Respiratory Diseases. *American Journal of Epidemiology*, 142(1): 15–22.

Stieb DM, Beveridge RC, Brook JR, Smith-Doiron M, Burnett RT, Dales RE, Beaulieu S, Judek S, Mamedov A. Air pollution, aeroallergens and cardiorespiratory emergency department visits in Saint John, Canada. *J Expo Anal Environ Epidemiol*. 2000 Sep-Oct;10(5):461-77.

Substantially more data are available pertaining to air pollution and hospital admissions in Canada relative to emergency department visits. We therefore elected to represent the effects of air pollution on respiratory emergency department visits using the results for respiratory hospital admissions scaled up in number based on the relative frequency of hospital admissions and emergency visits for these conditions. Thus, the change in frequency per unit air pollution was the same as for hospital admissions based on Burnett et al. (1995) i.e. 0.0754% (SE 0.0132%)

increase per 1 $\mu\text{g}/\text{m}^3$. The baseline rate of respiratory emergency visits is equal to the baseline rate of hospital admissions divided by 0.198, the proportion of visits resulting in hospital admission as reported by Stieb et al. (2000).

O₃ (May-Sep) and Respiratory Hospital Admissions

Burnett RT, Brook JR, Yung WT, Dales RE, Krewski D. 1997. Association between Ozone and Hospitalization for Respiratory Diseases in 16 Canadian Cities. *Environmental Research*, 72: 24–31.

Burnett et al. (1997) reported the results of a study of ozone and respiratory hospital admissions in 16 Canadian cities. Based on results from a Poisson regression model, which simultaneously adjusted for dewpoint temperature, CO, and coefficient of haze, they reported a relative risk of 1.024 ($p = 0.0258$) per 30 ppb daily one hour maximum ozone. Taking the natural logarithm of the relative risk and dividing by 30 yields a coefficient of 0.000791 (SE 0.000355) per ppb.

PM_{2.5} and Respiratory Hospital Admissions

Burnett RT, Dales R, Krewski D, Vincent R, Dann T, Brook JR. 1995. Associations between Ambient Particulate Sulphate and Admissions to Ontario Hospitals for Cardiac and Respiratory Diseases. *American Journal of Epidemiology*, 142(1): 15–22.

Burnett et al. (1995) reported a 3.5% (95%CI 2.3-4.7%) increase in respiratory hospital admissions per 13 $\mu\text{g}/\text{m}^3$ sulfate based on a linear regression model which also included ozone and temperature (table 4, row 2). Multiplying by the average ratio of sulfate to PM_{2.5} of 0.28 (Environment Canada), this equates to a 0.0754% (SE 0.0132%) increase per 1 $\mu\text{g}/\text{m}^3$ PM_{2.5}.

PM_{2.5} and Restricted Activity Days

Ostro, BD. 1987. Air Pollution and Morbidity Revisited: A Specification Test. *Journal of Environmental Economics and Management*, 14: 87-98.

Ostro BD and Rothschild S. 1989. Air Pollution and Acute Respiratory Morbidity: An Observational Study of Multiple Pollutants. *Environmental Research*, 50:238-247.

Chestnut et al. 1999: AIR QUALITY VALUATION MODEL (AQVM 3.0) REPORT 2: METHODOLOGY Final Report Prepared for: Environment Canada, Health Canada, September 3, 1999, by Lauraine G. Chestnut, David Mills, Shannon Ragland, Robert D. Rowe, of Stratus Consulting Inc., www.stratusconsulting.com.

Ostro (1987) reported an association between PM_{2.5} and reduced activity days (RADs) based on an analysis of data from the U.S. Health Interview Survey. They reported results by year for 1976-1981 based on a Poisson regression model (table III, column 2). We pooled these coefficients using a random effects model and obtained a pooled estimate of 0.00481 (SE 0.00101) per $\mu\text{g}/\text{m}^3$ PM_{2.5}. The baseline daily rate of RADs per person was 0.052 (Chestnut et al., 1999). Ostro and Rothschild (1989) also reported an analysis of PM_{2.5} and respiratory reduced activity days (RRADs), in which they adjusted for the simultaneous effects of ozone. The effects of PM_{2.5} were unaffected by this adjustment, thus we opted to use the results from their earlier analysis on the grounds that RADs are a more global outcome than the more narrowly defined RRADs.

Endpoint Valuations

All Mortality Endpoints

Lauraine G. Chestnut, Stratus Consulting Inc. and Paul De Civita, Policy Research Initiative. Economic Valuation of Mortality Risk Reduction: Review and Recommendations for Policy and Regulatory Analysis. Research Paper for PRI Project Regulatory Strategy, Canada, March 2009.

The default value provided in AQBAT for the value of a statistical life is \$6.5 million (2007 CDN dollars). It was derived from a literature review conducted by Stratus Consulting in 2009 (Chestnut, 2009). On advice from the review, low and high values were suggested to be \$3.5 million and \$9.5 million respectively which reflect possible ranges in the literature. It was also suggested that a probability weight be given to each value to account for the distribution of results in the literature. The authors suggest a weight of 50% for the mid value (\$6.5 million) and a weight of 25% for each of the low and high values.

The VSL literature review was based on revealed preference and stated preference studies that measured values associated with small changes in mortality risk. Based on the conclusions of the literature review, no adjustments to the VSL used in AQBAT are made due to age, size of mortality risk change, type of risk, health status or baseline risk.

Acute Respiratory Symptom Days

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. *Environ Health* 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

Adult Chronic Bronchitis Cases

Krupnick AJ and Cropper ML. 1992. The Effect of Information on Health Risk Valuations. *Journal of Risk and Uncertainty*, 5:29-48.

Viscusi WK, Magat WA, Huber J. 1991. Pricing Environmental Health Risks: Survey Assessments of Risk-Risk and Risk-Dollar Trade-offs for Chronic Bronchitis. *Journal of Environmental Economics and Management*, 21(1):32-51.

Chestnut et al. 1999: AIR QUALITY VALUATION MODEL (AQVM 3.0) REPORT 2: METHODOLOGY Final Report Prepared for: Environment Canada, Health Canada, September 3, 1999, by Lauraine G. Chestnut, David Mills, Shannon Ragland, Robert D. Rowe, of Stratus Consulting Inc., www.stratusconsulting.com.

Excerpt from AQVM 3.0 Methodology Report (Chestnut et al. 1999): "We use the reported median value from Viscusi et al. (1991) (\$457,000 1990 U.S. dollars) as the basis for our central estimate. For a low estimate for a severe case of chronic bronchitis we select the 20th percentile value of \$300,000 and for a high estimate we select the 80th percentile value of \$800,000 (1990 U.S. dollars) These values were converted to their 1996 Canadian dollar equivalents by first multiplying by the 1990 PPP index value of 1.22 and then inflating using the Canadian CPI values of 119.5 for 1990 and 135.7 for 1996. The mean severity rating reported for the Krupnick and Cropper sample is 6.5, based on the 0 to 13 scale. Using the elasticity at the mean of 1.16, this suggests that WTP for an average case is 58% lower than for a case at 13 on the scale. Using this to adjust the Viscusi et al. estimates, we get a central WTP estimate of \$266,000, a low of \$175,000, and a high of \$465,000 (1996 Canadian dollars), rounded to the nearest thousand, for an average case of chronic bronchitis."

Asthma Symptom Days

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. *Environ Health* 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5. Since in this case the 95% confidence interval was not symmetric around the point estimate, the distribution was characterized as triangular.

Cardiac Emergency Room Visits

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. *Environ Health* 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

Cardiac Hospital Admissions

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. *Environ Health* 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

NOTE:

According to Stieb et al. 2002, the probability and valuation of Cardiac Hospital Admissions (CHAs) are captured under valuation for Cardiac Emergency Room Visits (CERVs). To avoid double counting, valuation for CHAs is set to zero. The valuation for CERVs is applied to gross counts.

Child Acute Bronchitis Episodes

Krupnick AJ and Cropper ML. 1989. Valuing Chronic Morbidity Damages: Medical Costs, Labor Market Effects, and Individual Valuations. Final Report to U.S. EPA, Office of Policy Analysis.

Chestnut et al. 1999: AIR QUALITY VALUATION MODEL (AQVM 3.0) REPORT 2: METHODOLOGY. Final Report Prepared for: Environment Canada, Health Canada, September 3, 1999, by Lauraine G. Chestnut, David Mills, Shannon Ragland, Robert D. Rowe, of Stratus Consulting Inc., www.stratusconsulting.com.

Excerpt from AQVM 3.0 Methodology Report (Chestnut et al. 1999): "WTP estimates for bronchitis in children are not available. We therefore use the adjusted COI approach. Krupnick and Cropper (1989) report average annual medical treatment costs of \$42 in 1977 U.S. dollars for a child with bronchitis. This estimate is first inflated to its 1983 U.S. dollar equivalent using the U.S. medical consumer price index values of 57.0 for 1977 and 100.6 for 1983. The 1983 U.S. dollars value is then multiplied by the 1983 PPP index value of 1.24 and inflated using the Canadian medical care price index values of 85.1 for 1983 and 142.1 for 1996 to convert to the 1996 Canadian dollar equivalent of \$153. This treatment estimate is multiplied by the WTP/COI ratio of 2 to account for additional potential pain and suffering and activity losses. The central estimate is thus calculated as follows: Central \$/B/year = $\$153 \times \text{WTP/COI}$. Therefore, the central estimate is \$310. Applying a plus or minus 50% adjustment results in a low estimate of \$150 and a high estimate of \$460 (all values rounded to the nearest ten). These estimates are probably too low because they do not reflect any value for lost productivity during the time the children are ill. Monetary estimates for lost productivity because of illness for children are not readily available."

Elderly Cardiac Hospital Admissions

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. Environ Health 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

Minor Restricted Activity Days

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. Environ Health 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Since the value for reduced activity days in table 5 pertains to moderate levels of activity restriction, we used an alternative estimate based on the authors' sensitivity analysis, which employed a mild level of activity restriction. The valuation estimate in this case was \$22 (95%CI 4-40).

Respiratory Emergency Room Visits

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. Environ Health 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

Respiratory Hospital Admissions

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. Environ Health 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

NOTE:

According to Stieb et al. 2002, the probability and valuation of Respiratory Hospital Admissions (RHAs) are captured under valuation for Respiratory Emergency Room Visits (RERVs). To avoid double counting, valuation for RHAs is set to zero. The valuation for RERVs is applied to gross counts.

Restricted Activity Days

Stieb DM, De Civita P, Johnson FR, Manary M, Anis A, Beveridge RC, Judek S. Economic Evaluation of the Benefits of Reducing Acute Cardiorespiratory Morbidity Associated with Air Pollution. Environ Health 2002; 1:7.

In this paper, estimates are reported of the total valuation, comprising, where applicable, costs of illness, lost productivity, pain and suffering and averting expenditures. Total values are taken from table 5.

Appendix C – Analysis of Health Outcomes Related to Air Toxics Exposures (Benzene, 1,3-Butadiene, Formaldehyde, Acetaldehyde)

Cancer incidence (no threshold, gas phase air toxics)

Population annual cancer incidence is estimated using inhalation unit risks (IURs) as follows:

$$\text{Change in annual cancer incidence} = [(x_{FC} - x_{SQ}) \times \text{IUR} \times \text{population}] / 70$$

x_{FC} and x_{SQ} represent pollutant concentrations according to the targeted scenario (forecast) and reference scenario (status quo), respectively. An IUR value of 1×10^{-6} per $\mu\text{g}/\text{m}^3$ is interpreted as one excess cancer case per one million people if they are exposed daily and for a lifetime to $1 \mu\text{g}/\text{m}^3$ of a pollutant in air.

While CAREX Canada (2016)¹ and Public Health Ontario (2016)² weight these calculations by percent of time spent outdoors, application of the Hazardous Air Pollutant Exposure Model (HAPEM) suggests that ratios of personal exposures to ambient concentrations for acetaldehyde, benzene, 1,3-butadiene and formaldehyde are close to one, accounting for time activity patterns, penetration into microenvironments and amplification of average concentrations in the proximity of point and mobile sources (Ozkaynak et al. 2008)³. This was confirmed using data from the Canadian Human Activity Pattern Survey (CHAPS) 2 (Matz et al. 2014; results not shown)⁴. IURs are summarized below.

Impacts may be further characterized by multiplying the annual incidence by the number of Disability Adjusted Life Years (DALYs) per case. DALYs per case are summarized below.

Threshold carcinogens and non-cancer inhalation risks

The hazard associated with threshold carcinogens or non-cancer effects is estimated using reference concentrations (RfCs; also referred to as Reference Exposure Levels or RELs), which are estimates of continuous inhalation population exposures that are believed to be without appreciable risks of cancer or non-cancer effects over a lifetime. A hazard quotient (HQ) is calculated for each pollutant, by dividing the modelled concentration (x_{FC}) by its RfC as follows:

$$\text{HQ} = x_{FC} / \text{RfC}$$

A total hazard index (HI) is determined by summing the HQs for each pollutant at a location. The HI index is not a numerical risk estimate but a decision index. An HI of less than one is generally not expected to be a concern for health, whereas values above one indicate exposure concentrations that may present a health concern. Further, a HI of 0.2 or less for single pathways (in this case, inhalation) is considered negligible (Health Canada 2004)⁵. The US EPA notes that hazard quotients should be combined for pollutants that cause adverse effects by the same toxic mechanism. However, since detailed information on mechanisms is not available for most toxics, EPA aggregates the effects affecting the same target organ, regardless of the mechanism (USEPA 2017)⁶.

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Table A. Cancer unit risk per $\mu\text{g}/\text{m}^3$

Pollutant	Inhalation Unit Risk (per $\mu\text{g}/\text{m}^3$)
Acetaldehyde	Threshold carcinogen (use reference concentration of $280 \mu\text{g}/\text{m}^3$; see Table C)
Benzene	3.3×10^{-6} $3 \mu\text{g}/\text{m}^3$ for a 1×10^{-5} risk $\text{TC}_{05} = 15 \text{ mg}/\text{m}^3$
1,3-butadiene	5.88×10^{-6} $1.7 \mu\text{g}/\text{m}^3$ for 1×10^{-5} risk ($\text{TC}_{01} = 1.7 \text{ mg}/\text{m}^3$)
Formaldehyde	Threshold carcinogen (use reference concentration of $120 \mu\text{g}/\text{m}^3$; see Table C) Based on available data, the predicted additional risks of upper respiratory tract cancer for non-smokers, associated with an 80-year continuous exposure to levels of formaldehyde between 0.001 and 0.1 ppm (1.2 and $120 \mu\text{g}/\text{m}^3$), range from 2.3×10^{-10} to 2.7×10^{-8} , respectively (Environment Canada and Health Canada 2001).

TC_{xx} : substance-specific toxic concentration expected to produce a $xx\%$ increase in the cancer base rate.

Table B. DALY per cancer case.

Toxic	Cancer type	DALY per cancer case, in years
acetaldehyde	Nose	5.5
Benzene	Leukemia	13.7
formaldehyde	Nose and throat	5.5
1,3-butadiene	Leukemia	13.7

Source: Zhou et al. (2015).

Acetaldehyde, Benzene, Formaldehyde hazard quotient

Health Canada. Proposed residential indoor air quality guideline for acetaldehyde (<http://www.gazette.gc.ca/rp-pr/p1/2016/2016-12-17/pdf/g1-15051.pdf>)

OEHHA acute, 8 h and chronic reference exposure levels (RELs) as of June 2016 (www.oehha.ca.gov/air/allrels.html). Exposure averaging time for acute RELs is 1 h. Chronic RELs are designed to address continuous exposures for up to a lifetime; the exposure metric used is the annual average exposure.

Health Canada. Residential indoor air quality guideline for formaldehyde (<http://www.hc-sc.gc.ca/ewh-semt/pubs/air/formaldehyde/fact-info-eng.php>)

Table C. Reference concentrations, in $\mu\text{g}/\text{m}^3$, and target organ systems

Exposure	Acetaldehyde	Benzene	1,3-Butadiene	Formaldehyde
1 h	1 420 ^a Respiratory (bronchoconstriction)	n.a.	n.a.	123 ^d Ocular
Annual	280 ^a Respiratory (histological)	3 ^b Hematologic	1.7 ^c Cancer	50 ^d Respiratory (asthma)

^a Health Canada. Proposed residential indoor air quality guideline for acetaldehyde (<http://www.gazette.gc.ca/rp-pr/p1/2016/2016-12-17/pdf/g1-15051.pdf>)

^b OEHHA acute, 8 h and chronic reference exposure levels (RELs) as of June 2016 (www.oehha.ca.gov/air/allrels.html). Exposure averaging time for acute RELs is 1 h. Chronic RELs are designed to address continuous exposures for up to a lifetime; the exposure metric used is the annual average exposure.

^c Health Canada. 2004. Health-based guidance values for substances on the Second Priority Substances List (PSL2) (<https://www.canada.ca/en/health-canada/services/environmental-workplace-health/reports-publications/environmental-contaminants/health-based-guidance-values-substances-second-priority-substances-list.html>). Health Canada. 2017. Indoor air reference levels for chronic exposure to volatile organic compounds. (<https://www.canada.ca/en/health-canada/services/publications/healthy-living/indoor-air-reference-levels-chronic-exposure-volatile-organic-compounds.html>)

^d Health Canada. Residential indoor air quality guideline for formaldehyde (<https://www.canada.ca/en/health-canada/services/environmental-workplace-health/reports-publications/air-quality/formaldehyde-indoor-air-environment-workplace-health.html>)